

IntentFlow: Comprehensive Project Report

1. Executive Summary

IntentFlow is a state-of-the-art AI SEO Platform designed to bridge the visibility gap in the evolving landscape of AI-driven search. As users transition from traditional search engines like Google to AI chat providers (ChatGPT, Claude, Perplexity), SEO strategy must evolve to capture user intent from these conversational streams. IntentFlow provides a comprehensive monorepo ecosystem that captures, materializes, and analyzes AI search behavior into deterministic, actionable campaign data.

2. Technical Architecture

2.1 Backend Materialization Engine

The backend is a high-performance Node.js/Express service utilizing Prisma ORM with PostgreSQL. Its primary responsibility is the materialization of raw AI stream events into a hierarchical tree structure (Prompt -> Subquery -> Site/Evidence). It features a robust JWT-based RBAC system and uses BullMQ for distributed job orchestration.

2.2 Extension Interception Layer

A Manifest V3 Chrome extension serves as the primary data ingestion point. It intercepts live AI provider streams (ChatGPT, Claude, Perplexity) using MAIN-world content scripts and forwards normalized data to the backend. It also includes a side-panel UI for real-time visualization and campaign management.

2.3 Operator Web Dashboard

Built with React 19 and Vite, the web dashboard offers advanced data visualization using XYFlow for campaign trees and Recharts for engagement analytics. It provides administrative controls for user management, event tracking, and lead intelligence scoring.

IntentFlow: Comprehensive Project Report

3. Team Members & Detailed Contributions

Sibtain Ahmed Qureshi - Project Lead & Architect

- **Core Architecture:** Designed and implemented the monorepo structure, ensuring seamless interaction between backend, web, and extension surfaces.
- **Materialization Engine:** Developed the deterministic mapping logic that converts unstructured AI responses into structured relational data.
- **Database Infrastructure:** Architected the Prisma schema and led the migration to PostgreSQL for enterprise-grade scalability.

Abhishek Verma - Strategy & Systems Design

- **Lead Intelligence Framework:** Conceptualized and designed the lead intelligence module, defining scoring models and behavioral signal capture.
- **System Governance:** Established the project's 'Agent Context' and governance rules, enabling scalable AI-assisted development across the team.
- **Security & RBAC:** Designed the multi-tenant architecture and granular RBAC (Role-Based Access Control) framework for the platform.

Amogha Raj Sandur - DevOps & Quality Assurance

- **Project Governance:** Managed the Task workspace and CI/CD alignment, ensuring that implementation tasks are mapped to strategic milestones.
- **Code Quality:** Led refactoring efforts across the monorepo to maintain high modularity and technical debt reduction.
- **Documentation:** Architected the technical handoff documentation (context.md) and established standards for task-based progress tracking.

Mohammed Yaseen - Full-Stack Engineering

- **Mock Ecosystem:** Built the comprehensive Mock Server and testing environment for local-first development without provider dependencies.
- **Integration & Sync:** Led the cross-surface synchronization efforts, ensuring data consistency between the Extension side-panel and Web Dashboard.

IntentFlow: Comprehensive Project Report

- **Feature Development:** Implemented key onboarding flows and analytics event emitters across the platform surfaces.

Saksham - Backend & Data Pipelines

- **Data Pipelines:** Developed the integration bridges for SEMrush and Ahrefs, automating the collection of deep SEO insights.

- **Job Orchestration:** Optimized the BullMQ worker strategy for high-concurrency ingestion and enrichment tasks.

- **NLP Integration:** Implemented initial NLP-based intent extraction pipelines to categorize user queries by budget, location, and pain points.

Saumya - Frontend & UX Engineering

- **Visualization Systems:** Developed the custom XYFlow integration for deterministic campaign graphs, enabling interactive exploration of AI-search trees.

- **Compact Explorer:** Designed and implemented the 'Compact Tree Explorer' in the extension, significantly reducing cognitive load for operators.

- **Design System:** Maintained the shadcn/radix-based UI component library, ensuring aesthetic consistency and premium interaction patterns.

IntentFlow: Comprehensive Project Report

4. Future Scope & Roadmap

4.1 Server-Driven Replay & Automation

Transitioning from extension-based replay orchestration to a fully server-side execution model. This will allow for scheduled, background campaign refiring and monitoring without requiring active browser sessions.

4.2 Advanced Lead Scoring & CRM Integration

Implementing sophisticated machine learning models to refine lead scores based on historical interaction data. Planned automated exports to CRM platforms like Salesforce and HubSpot via custom webhook pipelines.

4.3 Enterprise Observability & Scaling

Integrating OpenTelemetry for end-to-end observability of data capture events. Scaling the Redis-based job queue to handle thousands of concurrent ingestion streams from enterprise pilot users.

4.4 Pilot Readiness & Commercial Launch

Finalizing the multi-tenant billing infrastructure and hardening security protocols for a public pilot launch. Focusing on SOC2 compliance readiness and high-availability deployment strategies.